ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 9 is divisible by 3 but not by 6	Counterexample for the converse.
2	**(d)** 6	6 is even.
3	**(b)** 4	$n = 4$ gives 121, which is composite.
4	**(d)** 12	List the factors of 72 in pairs; the count is 12 (even, so 72 is not a perfect square).
5	**(b)** right-angled	$33^2 + 56^2 = 4225$ and $65^2 = 4225$.
6	**(a)** $n(n+1)$ is even	One of two consecutive integers is even.
7	**(c)** 95	It equals 5×19 (number of terms times the middle term).
8	**(b)** 7	Only perfect squares have an odd number of factors.
9	**(d)** a 4 cm by 6 cm rectangle is not a square	Counterexample for the converse.
10	**(d)** 12	12 is divisible by 4 and 6 but not by 24.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Converse: If a number has an odd number of factors, then it is a perfect square. (1); it is true (1).	1+1
12	$n = 4$ (1); value 25 is composite (1).	1+1
13	3 factors (1); odd, so a perfect square (1).	1+1
14	Converse: If two sides of a triangle are equal, then the angles opposite them are equal. (1); it is true (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If the square of the longest side of a triangle equals the sum of the squares of the other two sides, then the triangle is right-angled. (1); The converse is also true (1); Both the proposition and its converse hold, so we may write the two as a single 'if and only if' statement (1).	1+1+1
16	A false statement needs only one counterexample (1); Take 18 (1); 18 is divisible by 3 and 9, not 27, so the statement is false (1).	1+1+1
17	Multiple of 18 is a multiple of 6 as $18 = 6 \times 3$; Converse: if n is a multiple of 6, then of 18 (1); Counterexample: 6 is a multiple of 6 but not of 18 (1); Converse is false (1).	1+1+1
	OR: Converse: If the diagonals of a parallelogram are equal, then it is a rectangle. (1); The converse is also true (1); Both the proposition and its converse hold, so we may write the two as a single 'if and only if' statement (1).	1+1+1
18	A false statement needs only one counterexample (1); Take 6 (1); 6 is even, so the statement is false (1).	1+1+1
	OR: Multiple of 24 is a multiple of 8 as $24 = 8 \times 3$; Converse: if n is a multiple of 8, then of 24 (1); Counterexample: 8 is a multiple of 8 but not of 24 (1); Converse is false (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Numbers $2n, 2n+2, \ldots, 2n+4$ (1); Sum $= 6n + 6 = 6(n+1)$ (1); For $n = 4$: 30 (1); $30 = 6 \times 5$ (2).	1+1+1+2
	OR: $47 - (4+7) = 36$ (1); $36 = 4 \times 9$ (1); General number $10a + b$ (1); $10a + b - (a+b) = 9a$ (1); A multiple of 9 (1).	1+1+1+1+1
20	Numbers $2n, 2n+2, \ldots, 2n+8$ (1); Sum $= 10n + 20 = 10(n+2)$ (1); For $n = 7$: 90 (1); $90 = 10 \times 9$ (2).	1+1+1+2
	OR: $63 - (6+3) = 54$ (1); $54 = 6 \times 9$ (1); General number $10a + b$ (1); $10a + b - (a+b) = 9a$ (1); A multiple of 9 (1).	1+1+1+1+1

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.
- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.

- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k+1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	$n = 3$ gives 119, which is composite.
2	**(d)** 18	List the factors of 180 in pairs; the count is 18 (even, so 180 is not a perfect square).
3	**(b)** right-angled	$20^2 + 99^2 = 10201$ and $101^2 = 10201$.
4	**(c)** $n^2 + n$ is even	$n^2 + n = n(n+1)$.
5	**(a)** 182	It equals 7×26 (number of terms times the middle term).
6	**(d)** 11	Only perfect squares have an odd number of factors.
7	**(a)** a triangle with sides $5, 5, 8$ is isosceles but not equilateral	Counterexample for the converse.
8	**(a)** $n = 4$	$2(4) + 1 = 9 = 3 \times 3$.
9	**(d)** 3	$n = 3$ gives 35, which is composite.
10	**(a)** 15	List the factors of 144 in pairs; the count is 15 (odd, so 144 is a perfect square).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$n = 10$ (1); value 121 is composite (1).	1+1
12	12 factors (1); even, so not a perfect square (1).	1+1
13	Converse: If two angles are equal, then they are vertically opposite. (1); it is false, counterexample: two angles of $60°$ in an equilateral triangle are equal but not vertically opposite (1).	1+1
14	$n = 3$ (1); value 9 is composite (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If a number is a multiple of 5, then it is a multiple of 15. (1); Counterexample: 20 is a multiple of 5 but not of 15 (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
16	A false statement needs only one counterexample (1); Take $n = 40$ (1); $1600 + 40 + 41 = 1681 = 41^2$, so the statement is false (1).	1+1+1
17	Multiple of 30 is a multiple of 10 as $30 = 10 \times 3$; Converse: if n is a multiple of 10, then of 30 (1); Counterexample: 10 is a multiple of 10 but not of 30 (1); Converse is false (1).	1+1+1
	OR: Converse: If a number is divisible by 4, then it is divisible by 12. (1); Counterexample: 8 is divisible by 4 but not by 12 (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
18	A false statement needs only one counterexample (1); Take 11 and 3 (1); $11 - 3 = 8$, so the statement is false (1).	1+1+1
	OR: Multiple of 36 is a multiple of 9 as $36 = 9 \times 4$; Converse: if n is a multiple of 9, then of 36 (1); Counterexample: 9 is a multiple of 9 but not of 36 (1); Converse is false (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Numbers $2n, 2n+2, \ldots, 2n+12$ (1); Sum $= 14n + 42 = 14(n+3)$ (1); For $n = 3$: 84 (1); $84 = 14 \times 6$ (2).	1+1+1+2
	OR: $85 - (8 + 5) = 72$ (1); $72 = 8 \times 9$ (1); General number $10a + b$ (1); $10a + b - (a + b) = 9a$ (1); A multiple of 9 (1).	1+1+1+1+1
20	Numbers $2n, 2n+2, \ldots, 2n+4$ (1); Sum $= 6n + 6 = 6(n+1)$ (1); For $n = 10$: 66 (1); $66 = 6 \times 11$ (2).	1+1+1+2
	OR: $92 - (9 + 2) = 81$ (1); $81 = 9 \times 9$ (1); General number $10a + b$ (1); $10a + b - (a + b) = 9a$ (1); A multiple of 9 (1).	1+1+1+1+1

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.
- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** right-angled	$39^2 + 80^2 = 7921$ and $89^2 = 7921$.
2	**(d)** $2n + 1$ is odd	$2n$ is even.
3	**(c)** 126	It equals 3×42 (number of terms times the middle term).
4	**(d)** 14	Only perfect squares have an odd number of factors.
5	**(d)** $x = -3$ gives $x^2 = 9$ but $x \neq 3$	Counterexample for the converse.
6	**(b)** 11 and 3	$11 - 3 = 8$.
7	**(a)** 3	$n = 3$ gives 65, which is composite.
8	**(b)** 16	List the factors of 210 in pairs; the count is 16 (even, so 210 is not a perfect square).
9	**(c)** not right-angled	$45^2 + 60^2 = 5625$ and $76^2 = 5776$.
10	**(d)** $n^3 - n$ is divisible by 2	$n^3 - n = (n-1)n(n+1)$ has an even factor.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3 factors (1); odd, so a perfect square (1).	1+1
12	Converse: If the sum of the digits of a number is divisible by 9, then the number is divisible by 9. (1); it is true (1).	1+1
13	$n = 5$ (1); value 25 is composite (1).	1+1
14	12 factors (1); even, so not a perfect square (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If all angles of a triangle are $60°$, then the triangle is equilateral. (1); The converse is also true (1); Both the proposition and its converse hold, so we may write the two as a single 'if and only if' statement (1).	1+1+1
16	A false statement needs only one counterexample (1); Take 12 (1); 12 is divisible by 4 and 6 but not by 24, so the statement is false (1).	1+1+1
17	Multiple of 28 is a multiple of 7 as $28 = 7 \times 4$; Converse: if n is a multiple of 7, then of 28 (1); Counterexample: 7 is a multiple of 7 but not of 28 (1); Converse is false (1).	1+1+1
	OR: Converse: If the sum of the digits of a number is divisible by 9, then the number is divisible by 9. (1); The converse is also true (1); Both the proposition and its converse hold, so we may write the two as a single 'if and only if' statement (1).	1+1+1
18	A false statement needs only one counterexample (1); Take $\sqrt{2} \times \sqrt{2}$ (1); $\sqrt{2} \times \sqrt{2} = 2$ is rational, so the statement is false (1).	1+1+1
	OR: Multiple of 45 is a multiple of 15 as $45 = 15 \times 3$; Converse: if n is a multiple of 15, then of 45 (1); Counterexample: 15 is a multiple of 15 but not of 45 (1); Converse is false (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Numbers $2n, 2n+2, \ldots, 2n+8$ (1); Sum $= 10n + 20 = 10(n+2)$ (1); For $n = 12$: 140 (1); $140 = 10 \times 14$ (2).	1+1+1+2
	OR: $38 - (3 + 8) = 27$ (1); $27 = 3 \times 9$ (1); General number $10a + b$ (1); $10a + b - (a + b) = 9a$ (1); A multiple of 9 (1).	1+1+1+1+1
20	Numbers $2n, 2n+2, \ldots, 2n+16$ (1); Sum $= 18n + 72 = 18(n+4)$ (1); For $n = 2$: 108 (1); $108 = 18 \times 6$ (2).	1+1+1+2
	OR: $71 - (7 + 1) = 63$ (1); $63 = 7 \times 9$ (1); General number $10a + b$ (1); $10a + b - (a + b) = 9a$ (1); A multiple of 9 (1).	1+1+1+1+1

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.

- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND